Zero to Hero questions on Strings in C

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1 String Programming Questions in C (Grouped by Concepts)

The goal is to solve problems that naturally combine multiple string functions and operations rather than practicing each function separately.

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2 Level 1: Basic String Handling

2.0.1 1. Student Information Formatter

Problem: Accept a student's first name and last name separately. Combine them into a full name and display its length.

Concepts:

- Input
- `strcat()`
- `strlen()`

Hint: Store first and last names in separate arrays and concatenate them with a space.

2.0.2 2. Username Validator

Problem: Accept a username and check whether it contains at least 6 characters.

Concepts:

- `strlen()`
- Decision making

Hint: Length should be greater than or equal to 6.

2.0.3 3. String Copy and Compare

Problem: Copy a string into another variable and verify whether the copy was successful.

Concepts:

- `strcpy()`

- `strcmp()`

Hint: Comparison should return 0.

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3 Level 2: Searching and Counting

3.0.1 4. Character Analyzer

Problem: Count vowels, consonants, digits, spaces, and special characters in a string.

Concepts:

- Loops
- Character classification

Hint: Process each character individually.

3.0.2 5. Word Search Tool

Problem: Check whether a given word exists inside a sentence.

Concepts:

- `strstr()`

Hint: If return value is not NULL, word exists.

3.0.3 6. Character Frequency Report

Problem: Display frequency of every character appearing in a string.

Concepts:

- Arrays
- Nested loops

Hint: ASCII table has 256 possible characters.

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4 Level 3: String Transformation

4.0.1 7. Reverse and Palindrome Checker

Problem: Reverse a string and determine whether it is a palindrome.

Concepts:

- Reverse
- `strcmp()`

Hint: Compare original and reversed strings.

4.0.2 8. Case Converter

Problem: Convert lowercase letters to uppercase and uppercase letters to lowercase.

Concepts:

- `toupper()`
- `tolower()`

Hint: Use `<ctype.h>`.

4.0.3 9. Remove Duplicate Characters

Problem: Remove repeated characters from a string.

Concepts:

- Nested loops
- String shifting

Hint: Shift remaining characters left.

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5 Level 4: Combining Functions

5.0.1 10. Sentence Statistics Generator

Problem: Display:

- Total characters
- Total words
- Total vowels
- Total consonants
- Longest word

Concepts:

- `strlen()`
- `strtok()`

Hint: Tokenize words one by one.

5.0.2 11. Password Strength Checker

Problem: Determine whether a password contains:

- Uppercase

- Lowercase
- Digit
- Special character
- Minimum length

Concepts:

- `strlen()`
- `ctype.h`

Hint: Use flags for each condition.

5.0.3 12. Email Validator

Problem: Check whether an email contains:

- '@'
- '.'
- Valid placement

Concepts:

- `strchr()`
- `strrchr()`

Hint: '.' should appear after '@'.

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6 Level 5: Advanced Manipulation

6.0.1 13. Substring Replace Utility

Problem: Replace all occurrences of one word with another.

Concepts:

- `strstr()`
- Copying

Hint: Search repeatedly until NULL.

6.0.2 14. Word Sorter

Problem: Sort words in a sentence alphabetically.

Concepts:

- `strtok()`
- `strcmp()`
- `strcpy()`

Hint: Store tokens in 2D array.

6.0.3 15. Anagram Checker

Problem: Check whether two strings are anagrams.

Concepts:

- Sorting
- Comparison

Hint: Sort both strings then compare.

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7 Level 6: Real Interview-Level Questions

7.0.1 16. Mini Search Engine

Problem: Accept a paragraph and a keyword. Display:

- Number of occurrences
- First occurrence
- Last occurrence

Concepts:

- `strstr()`
- Pointer arithmetic

Hint: Continue searching from previous match + 1.

7.0.2 17. Text Compression

Problem: Convert:

`aaabbcccc`

into:

`a3b2c4`

Concepts:

- Frequency counting
- String building

Hint: Count consecutive characters.

7.0.3 18. Caesar Cipher

Problem: Encrypt and decrypt a sentence.

Concepts:

- Character manipulation
- Uppercase/lowercase handling

Hint: Shift alphabet positions.

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8 Level 7: Function Group Challenges

8.1 Challenge A: Use `strlen() + strcpy() + strcmp()`

Problem: Create a backup system that:

1. Accepts a string.
2. Creates a backup copy.
3. Verifies copy integrity.
4. Displays length of both strings.

Hint: All lengths and contents should match.

8.2 Challenge B: Use `strcat() + strlen() + strstr()`

Problem: Merge two paragraphs and check whether a given keyword exists.

Hint: Concatenate first, search later.

8.3 Challenge C: Use `strchr() + strrchr() + strlen()`

Problem: For a given sentence display:

- First occurrence of a character
- Last occurrence of a character
- Distance between them

Hint: Subtract pointers.

8.4 Challenge D: Use `strtok() + strcmp() + strcpy()`

Problem: Find the lexicographically smallest and largest word in a sentence.

Hint: Tokenize and compare words one by one.

9 Level 8: One Mega Project (Uses Almost Everything)

9.0.1 Text Processing System

Problem:

Accept a paragraph and perform:

1. Length calculation
2. Word count
3. Vowel count
4. Consonant count
5. Digit count
6. Space count
7. Reverse text
8. Palindrome check
9. Uppercase conversion
10. Lowercase conversion
11. Search a word
12. Replace a word
13. Remove duplicates
14. Sort words
15. Encrypt text
16. Decrypt text
17. Save backup copy
18. Verify backup using comparison

Functions/Concepts Covered:

- `strlen()`
- `strcpy()`
- `strcmp()`
- `strcat()`
- `strchr()`
- `strrchr()`
- `strstr()`
- `strtok()`
- `toupper()`
- `tolower()`
- Arrays
- Loops
- Pointers
- Character manipulation

Hint: Implement each feature as a separate function and create a menu-driven program.

This final project effectively covers nearly every string operation commonly asked in C programming courses, interviews, competitive coding, and lab examinations.